- The balances of all accounts are fetched again. The result of the auction contains the address of the winner (ACC3, in this case) and the winning bid amount (BID3). The ending balances are checked. The account balances of bidders who lost are close to the starting balances because only Txs cost is used up. The winner of the auction balance is reduced by the bid amount plus any Tx cost.

In this fully integrated run, the script exhaustively exercises the functions of the blind auction in a single test and verifies that the smart contract works as expected.

10.6 Retrospective

In all three cases, the smart contract testing was successful. This situation may not be the case in the development stages as a team is developing the functions of the smart contract. Change some number value in any of the smart contracts discussed in this chapter, and you'll get one or more tests failing (X instead of a check in the test output). I recommend that you try this now.

In a test-driven development, the test script is written first; then the smart contract is developed to meet the requirements specified in the test. Testing can be used as a means of maintaining the integrity of a codebase when multiple developers are contributing to its parts. In this scenario, test scripts can be used to make sure that the code committed meets the requirements.